SOFTWARE ARCHITECTURE AND DESIGN SPECIFICATION
Collaborative Study Material & Quiz Platform

SRN	Team No.	Name
PES2UG24CS003	16	A Sheiman Joshi
PES2UG24CS033	16	Aditya Raj
PES2UG24CS056	16	Amrit Raj Anmol
PES2UG24CS037	16	Ahan Charan

Version 1.0
Date 14 September 2026
Status: Draft

Revision History

Version	Date	Author	Change Summary
1.0	14 September 2026	Team 16	Initial Software Architecture and Design Specification

Approvals

Role	Name	Signature / Date
Course Coordinator	_____	_____
Project Evaluator	_____	_____

1. Introduction

1.1 Purpose

This Software Architecture and Design Specification defines the architecture and design of the Collaborative Study Material & Quiz Platform. It translates the requirements established in the SRS into a technology neutral architecture, component structure, interaction design, interface design, security approach, risks, and design decisions.

1.2 Scope

The system is a web application providing username and password authentication, role based access, study material contribution and discovery, collaborative editing, version history, comments, ratings, quiz creation, timed and randomized quiz attempts, automatic evaluation, instant results, leaderboards, quiz history, and administrative management.

1.3 Audience

- Project developers and student team members
- QA and test engineers and evaluators
- Course instructors and project reviewers
- System and design documentation reviewers

1.4 Definitions

Term	Definition
SAD	Software Architecture and Design Specification
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
RBAC	Role Based Access Control
Version	Saved state of study material retained for history or recovery
Quiz Creator	Authorized user who creates and manages quizzes
Content Contributor	Authorized user who uploads or collaboratively edits study material

2. Document Overview

2.1 How to Use This Document

This document describes the selected architecture, components, interactions, sequence flows, API contracts, security architecture, risks, traceability, and design considerations. It is intended to be read together with the project SRS.

2.2 Related Documents

Document	Purpose
Software Requirements Specification	Defines functional requirements, non functional requirements, security requirements, acceptance criteria, and traceability.
Software Architecture and Design Specification	Defines how the requirements are organized into architectural and design elements.

3. Architecture

3.1 Goals and Constraints

Area	Details
Goals	Clear separation of concerns, maintainability, security, data integrity, acceptable response time, and suitability for the academic project.
Web application	The system shall operate through a modern web browser.
Authentication	Authentication shall use username and password.
Team constraint	The implementation must remain achievable for a four member academic team.
Technology	The SRS intentionally leaves the technology stack open until design and implementation.
Storage	Persistent storage is required for users, content, quizzes, attempts, and uploaded documents.
Security	Protected operations shall use server side authorization and sensitive communication shall use HTTPS and TLS.

3.2 Stakeholders and Concerns

Stakeholder	Primary Concerns
Student / User	Material discovery, reliable downloads, quiz attempts, immediate results, performance history, and secure access.
Content Contributor / Quiz Creator	Content creation, collaboration, version history, quiz creation, and permissions.
Administrator	User management, content moderation, audit information, security, and configuration.
Project Developers	Modularity, documented interfaces, testability, maintainability, and achievable scope.
QA and Evaluators	Functional correctness, measurable quality attributes, traceability, and security verification.

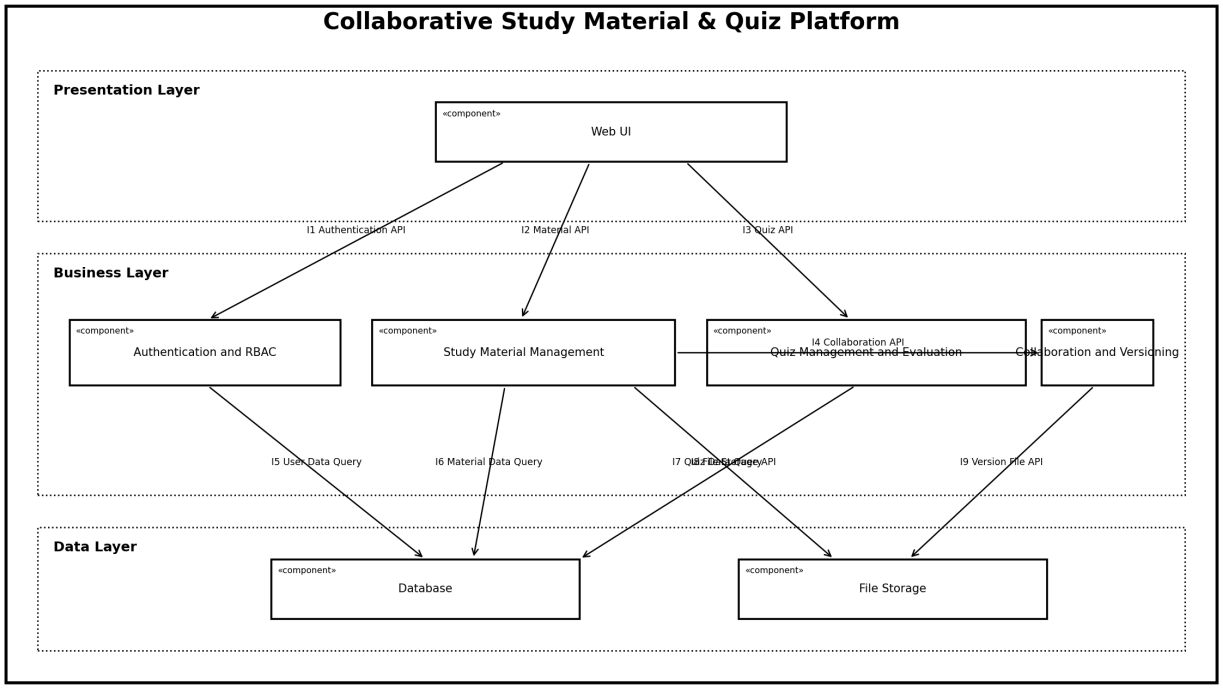
3.3 Chosen Architecture Pattern and Rationale

Layered Architecture is selected because the SRS naturally separates presentation, business processing, and persistent data services. The Web UI forms the Presentation Layer. Authentication and RBAC, Study Material Management, Collaboration and Versioning, and Quiz Management and Evaluation form the Business Layer. Database and File Storage form the Data Layer.

This architecture supports the four member academic scope without introducing the deployment and operational complexity of Microservices Architecture. It also supports the SRS maintainability requirement by keeping responsibilities separated and interfaces documented.

3.4 Component UML Diagram

The component model contains seven major components derived from the SRS responsibilities and external interface requirements.



Layered component architecture.

3.5 Component Descriptions

Component	Layer	Description
Web UI	Presentation	Login, dashboard, material pages, material editor, quiz builder, quiz attempt interface, result display, and administrator interface.
Authentication and RBAC	Business	Username and password authentication, sessions, and role based permissions.
Study Material Management	Business	Creation, upload, organization, search, filtering, download, editing, deletion, ratings, comments, and discussions.
Collaboration and Versioning	Business	Controlled collaborative updates and version history.
Quiz Management and Evaluation	Business	Quiz creation, question types, timing, randomization, attempts, evaluation, results, history, and leaderboard.
Database	Data	User, role, material, version, comment, rating, quiz, attempt, score, and audit records.
File Storage	Data	Uploaded study documents and version files.

3.6 Technology Stack and Data Stores

The SRS does not fix a technology stack. This SAD therefore does not invent a framework, programming language, database product, or hosting provider. During implementation planning, the team shall select a suitable web frontend, backend runtime, relational or document oriented database, file storage solution, and deployment environment. The selected environment shall provide HTTPS and TLS.

3.7 Risks and Mitigations

Risk	Mitigation
Unauthorized access	Centralize authentication and enforce server side authorization for protected operations.
Unsafe uploads	Validate and sanitize user supplied content and uploaded file metadata.
Collaborative data inconsistency	Use controlled updates and preserve each committed revision in version history.
Manipulated quiz scores	Evaluate quizzes on the server and protect authoritative result records.
Performance degradation	Measure response time and resource usage against the SRS criteria.

3.8 Traceability to Requirements

SRS Requirements	Architecture or Design Area
CSQP-F-001 and CSQP-F-002	Authentication and RBAC; Web UI
CSQP-F-003 through CSQP-F-010	Study Material Management; File Storage; Database
CSQP-F-011 and CSQP-F-012	Collaboration and Versioning; Database; File Storage
CSQP-F-013 through CSQP-F-019	Quiz Management and Evaluation; Database; Web UI
CSQP-F-020 through CSQP-F-022	Authentication and RBAC; Study Material Management; Database
CSQP-NF-001	Layered separation and dedicated application and data components
CSQP-NF-002	Deployment and operational monitoring
CSQP-NF-003	Web UI design
CSQP-NF-004	Database and controlled persistence
CSQP-NF-005	Layered modular component structure and documented interfaces
CSQP-NF-006	Application and data components for the configured workload
CSQP-SR-001 through CSQP-SR-006	Authentication, secure communication, validation, session handling, and server side result protection

3.9 Security Architecture

Security is organized as a cross cutting concern with Authentication and RBAC acting as the main authorization boundary. Sensitive communication uses HTTPS and TLS. Passwords are stored only as secure one way hashes. Protected material, quiz, administration, and user management operations require server side authorization. User supplied content and uploaded file metadata are validated and sanitized. Sessions are invalidated according to the configured logout and timeout policy. Quiz scores and authoritative results are calculated and stored on the server.

Security Objective	Architectural Control
SO-01 Confidentiality	Authentication, role based authorization, password hashing, HTTPS and TLS, and controlled data access.
SO-02 Integrity and Accountability	Server side authorization, server side evaluation, versioning, data integrity controls, and audit logging.

4. DESIGN

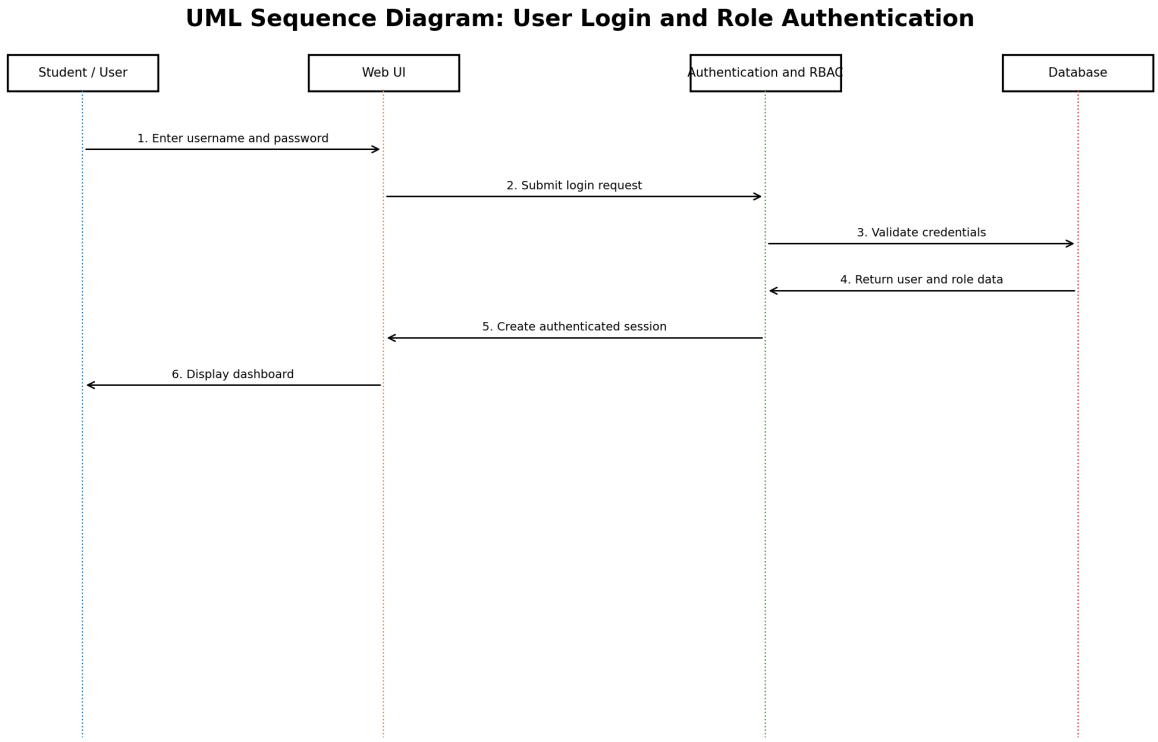
4.1 Design Overview

The design follows the selected layered architecture. The Web UI handles presentation and interaction. Business components apply application rules and authorization. Data components provide persistence for structured records and uploaded files.

4.2 UML Sequence Diagrams

The two sequence flows below are specific to the SRS and cover authentication and a timed quiz attempt with automatic evaluation and instant result display.

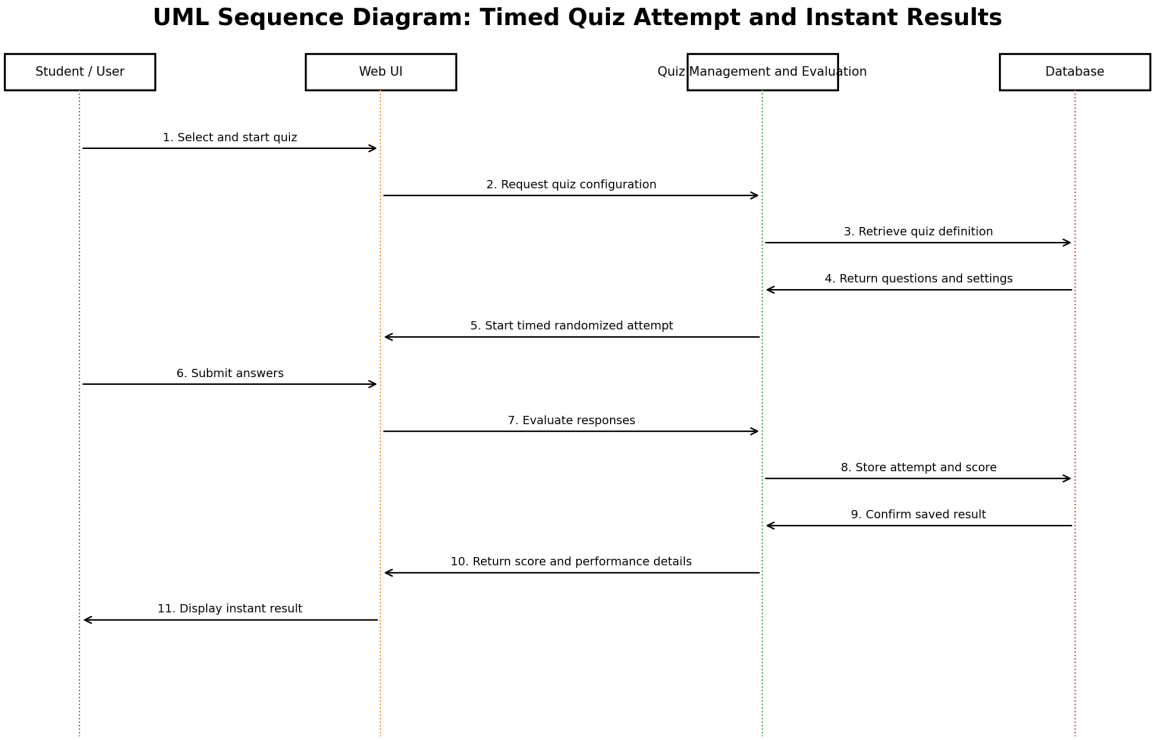
4.2.1 User Login and Role Authentication



Username and password authentication sequence.

- The user enters username and password through the Web UI.
- The Web UI submits the credentials to Authentication and RBAC.
- Authentication and RBAC validates the credentials using the Database.
- An authenticated session and user role are established.
- The Web UI displays the permitted dashboard.

4.2.2 Timed Quiz Attempt and Instant Results



Timed quiz, evaluation, and result sequence.

- The user selects and starts a quiz.
- The quiz component retrieves its definition and timing and randomization settings.
- The attempt begins with the configured timer and question order.
- The user submits answers.
- The quiz component evaluates responses, stores the attempt and score, and returns the result.
- The Web UI displays the score and performance details when instant results are enabled.

4.3 API Design

The following are logical technology neutral interfaces. Exact endpoint paths, serialization, session mechanism, and framework details shall be finalized during implementation.

API	Method	Request	Response	Representative Errors
Authentication API	POST	username, password	authenticated status, user role, session result	Invalid credentials, validation failure
Material API	GET, POST, PUT, DELETE	material metadata, content, search criteria, update data	material record, search results, operation status	Unauthorized, invalid input, file validation failure, resource not found
Quiz API	GET, POST	quiz identifier, quiz configuration, answers, submission data	quiz definition, attempt status, score, performance details	Unauthorized, unavailable quiz, invalid submission, evaluation failure

4.4 Error Handling, Logging and Monitoring

- Return clear and consistent error responses.
- Do not expose passwords or other sensitive information in logs or error messages.
- Record important user and administrative actions in the audit log.
- Monitor response time, availability, authentication failures, material failures, and quiz evaluation failures.
- Protect operational logs using appropriate access controls.

4.5 UX Design

Screen	Key Elements
Login	Username, password, authentication feedback
Dashboard	Subjects, topics, available material, recent content, quiz activity
Material Page	Metadata, preview or download, comments, ratings, version information
Material Editor	Create or edit content, save and update controls, collaboration controls
Quiz Builder	Question type, answer configuration, duration, randomization
Quiz Attempt	Question navigation, timer where applicable, submission confirmation, result
Administrator Interface	User management, content moderation, platform activity and configuration

4.6 Open Issues and Next Steps

Open Issue	Next Step
Technology stack not fixed	Select frontend, backend, database, file storage, and deployment environment.
Exact API contract not fixed	Finalize endpoint paths, request and response schemas, session mechanism, and error codes.
Concurrent editing strategy not fixed	Define update and conflict handling during detailed implementation design.
Monitoring details not fixed	Select monitoring metrics, log retention approach, and availability measurement.
Deployment environment not fixed	Choose an environment supporting HTTPS, persistent storage, and the configured workload.

5. Appendices

5.1 Glossary

Term	Meaning
API	Application Programming Interface
RBAC	Role Based Access Control
SRS	Software Requirements Specification
SAD	Software Architecture and Design Specification
HTTPS	HTTP communication protected by TLS
TLS	Transport Layer Security
Audit Log	Record of important user and administrative actions
Version History	Stored sequence of saved states of collaboratively edited material

5.2 References

- Collaborative Study Material & Quiz Platform Software Requirements Specification, Team 16, Version 1.0, dated 09 September 2026.
- Software Architecture and Design Specification template supplied for the project.

5.3 Tools

The architecture and sequence diagrams are supplied in editable draw.io format. The document is supplied in Microsoft Word format. Final implementation tools remain open until the technology stack is selected.
